

Renaissance Inspired Shepherd's Sundial

By ex_scopus in Outside > Backyard  5,300  68  23  Featured  Winner

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Purpose of the project

I sought a portable sundial for my loggia, suitable for use on sunny days while enjoying a cup of coffee. My preference was for a model that does not require a compass for alignment, can be easily relocated, and allows for tactile interaction and appreciation.

I chose to build a cylindrical sundial, also called a shepherd's or pillar sundial. This portable device, used from Antiquity, tells time by the sun's altitude, requiring only the current date and no compass.

Historical Significance

The shepherd's, pillar or cylindrical sundial has held considerable importance throughout history, symbolising both practical timekeeping and intellectual sophistication.

Project inspiration

My inspiration - Renaissance (1400 – 1600) designs of the shepherd's sundial. During the Renaissance, these sundials were celebrated for their artistic and scientific merit, frequently appearing in prominent works of art and literature.

The most iconic artistic representation of the cylindrical dial is in Hans Holbein's the Younger 1533 masterpiece, "The Ambassadors" ([Hans Holbein the Younger - The Ambassadors - Google Art Project - The Ambassadors \(Holbein\) - Wikipedia](#)).

The dial sits on the top shelf, which contains "heavenly" objects like celestial globes and quadrants, contrasting with the "earthly" items on the lower shelf. This specific dial features a bell-shaped cap and is decorated with the zodiacal signs for Aries and Virgo (The Ambassadors - Shepherd's dial - The Ambassadors (Holbein) - Wikipedia).

In 1528, Holbein painted a portrait of Nicolaus Kratzer, the astronomer and dial-maker to King Henry VIII. The painting includes a cylinder dial identical in style to the one in The Ambassadors, reflecting Kratzer's role as a technical advisor to Holbein on scientific instruments.

Principle of Operation

The shepherd's or pillar sundial functions by measuring the sun's altitude rather than its azimuth. This allows the user to find the time without a magnetic compass, provided the current date is known.

1. **Date Alignment:** The user rotates the top of the dial (the gnomon support) until the gnomon is positioned directly over the vertical line corresponding to the current date or Zodiacal sign.
2. **Gnomon Extension:** The gnomon is extended horizontally, perpendicular to the cylinder, mounted on rotatable top.
3. **Solar Alignment:** The user rotates the whole cylinder directly toward the sun. This is achieved when the gnomon casts a narrow, vertical shadow straight down the side of the cylinder.
4. **Reading the Time:** The time is indicated by the position of the tip of the shadow. The point where the shadow's end falls among curved hour lines reveals the time.

Supplies



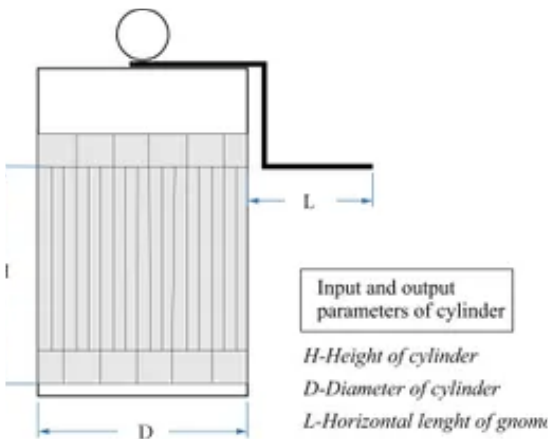
1. Cylinder – I used a metal coffee can, but any cylinder works.
2. Scissors;
3. Multitool or wire cutter;
4. Ruler;
5. Pencil;
6. Few sheets of printer paper;
7. Delicate painter's tape (Usually rose or violet colour. Easy to remove, no residue left.);
8. Small magnet and piece of wire for gnomon;
9. Small wooden cabinet handle;
10. Paper glue;
11. Universal glue (optional, to glue gnomon to magnet and gnomon to cover);
12. Double sided adhesive tape (to glue gnomon to cover).

Step 1: Measure a Cylinder



With delicate painter's tape (and ruler) measure circumference and height of cylinder (in millimetres). Calculate diameter of cylinder by dividing circumference by 3,14159 (π).

Step 2: Design the „raw” Dial



I wrote a Python script designed for the generation of shepherd's or cylindrical sundial, historically referred to as pillar dial, cylinder, or chilindre.

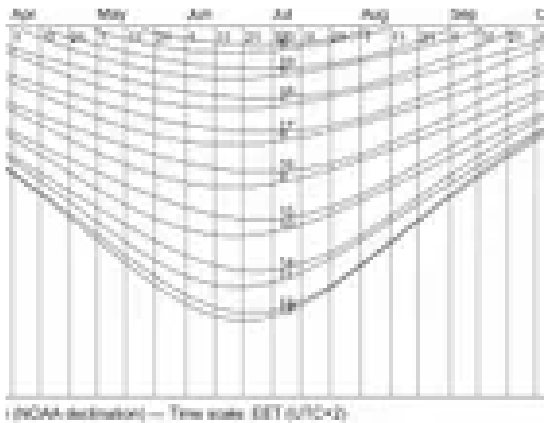
The script takes latitude, time zone, cylinder diameter, and height (all in millimetres) as inputs to calculate the required gnomon and generates an editable Inkscape .svg dial. Simply edit the script with your parameters and save it.

This version replaces the traditional declination (and optionally EOT) approximation with NOAA's 'General Solar Position Calculations' declination model (NOAA GML solar equations (declination + equation of time) from <https://gml.noaa.gov/grad/solcalc/solareqns.PDF>)

Script designs the sundial for standard time (“clock time”) instead of local time. For the Daylight Time Saving Time (“Summer time”) you must add 1 hour.

I placed the script I created in my Github: [Github - zodiakas/sundial: Sundial design scripts · Github](#)

Step 3: Design the Final Dial



I ran a Python script on my Raspberry Pi 5 single board computer, and then used the free Inkscape vector editor to modify the resulting .svg file. I made these modifications:

- Changed fonts and font sizes;
- Changed line styles of some lines;
- Removed shadow lines before 6:00 and after 19:00;
- Added ribbon with the Zodiac signs at the bottom of the dial;
- Added text boxes for mottos.

Step 4: Write Inspiring Mottos

A Renaissance sundial was not just a device for telling time. It was also a moral and literary object: a thing that measured hours while prompting reflection on life, death, work, virtue, and the heavens. Sundials were “both a way of telling time and a focal point for reflecting on its nature.”

Renaissance sundial mottos are a fascinating mix of timekeeping, poetry, moral reflection, and visual culture. In Europe, sundials began to carry mottoes about the passage of time, mortality, proper conduct, and the use of the day in the 1500s, and by the early modern period the inscription had become almost an expected finishing touch on many dials rather than an occasional extra. They turned a scientific instrument into a humanist emblem of time—half clock, half poem, half sermon. Latin was the prestige language, and many mottos were either quotations from or inspired by Roman authors such as Virgil, Horace, and Martial. For example, “Tempus fugit” (“Time flees”) is explicitly linked to Virgil’s *Georgics*.

I chose two uplifting mottos for my sundial, which I will enjoy and reflect on whenever I see it:

1. Semper Sol in corde tuo luceat (May the Sun always shine in your heart).
2. Horas non numero nisi serenas. (I do not count the hours unless they are sunny).

Step 5: Montage of the Instrument



1. Print the final dial .svg at 1:1 scale using Inkscape.
2. Cut out the dial, leaving a small edge margin to glue.
3. Test fit on the cylinder; adjust the .svg if necessary.
4. Glue the dial into a tube shape and slide it onto the cylinder.
5. Attach an adhesive magnet to the wooden cabinet handle.
6. Make a gnomon from wire, fix it to a small magnet and cabinet handle, secure with drop of glue.
7. Draw circle around the cover of coffee can on paper, find and mark the cylinder's centre using chords and bisectors, transfer the centre mark to the coffee can cover.
8. Place the gnomon (wire and magnet attached to the cabinet handle) in the cover's centre.
9. Bend the gnomon wire as needed, adjust and make easily removable marks on cover to permanently place gnomon.
10. Follow the marks and fix gnomon to cover with double sided adhesive tape or drop of glue.

11. Adjust gnomon length (cut it if needed, leave some extra for calibration).

Step 6: Use the Sundial



1. Rotate cover with gnomon to desired date on dial.
2. Orient the sundial to the direction of the Sun, while keeping it on flat and horizontal surface. Align the gnomon's shadow vertically. Tip: First, rotate the sundial so, that the knob's shadow matches the line of the gnomon, then adjust until the gnomon's shadow is perfectly vertical (look at the dial's lines).
3. Read the time from the shadow cast by gnomon.
4. If your country is observing Daylight Saving Time on this date, add one hour.
5. Compare the present time displayed on both your sundial and conventional clock. Calibrate the sundial by finalising length of the gnomon (optional).

Sundial works on a window ceiling as well as outside (preferred use), but refraction in the glass will reduce its accuracy (in my case +/- 0.5h).

Tip: Print the dial scale on coloured paper and switch it up to match your mood.

Enjoy! May the Sun always shine in your heart!